Utkarsh Kumar

Software Engineer | Data Structures & Algorithms | Systems Design | Machine Learning
+91 7352717183 | unofficialutkarsh.06@gmail.com | GitHub | LinkedIn | LeetCode (350+ solved) | Portfolio

Summary

Third-year Computer Science student with 3+ years of programming experience and strong foundations in data structures, algorithms, and object-oriented design (Python, C++, Java) – 350+ problems solved on LeetCode. Hands-on experience designing, testing, and shipping production software across backend systems, distributed pipelines, and machine learning applications, working closely with cross-functional teammates from requirements through deployment and monitoring. Published IGARSS 2026 researcher and multi-hackathon winner, quick to learn new engineering methods and tools, and comfortable incorporating peer and industry feedback to raise code quality in fast-paced, collaborative environments.

Education

Manipal Institute of Technology, Bengaluru GPA: 9.3/10
B.Tech in Computer Science (Data Science) 2024 – 2028

Indian Institute of Technology, Guwahati GPA: 8.3/10
B.Sc Honours in Artificial Intelligence and Data Science (pursued concurrently) 2024 – 2028

Technical Skills

Languages (OOP): Python, C++, Java, JavaScript, TypeScript, SQL

Software Engineering: Data Structures & Algorithms, Object-Oriented Design, System Design, Software Design Patterns, Unit & Integration Testing, Debugging, Code Review, Agile/Scrum, CI/CD, Version Control (Git)

Backend / Systems: FastAPI, Flask, Node.js, Next.js, React, PostgreSQL, Redis, MongoDB, SQLAlchemy, Pydantic, Docker, Kubernetes, Microservices, REST APIs, WebSockets, GraphQL, Distributed Systems

Cloud / DevOps / Observability: Azure (AKS, AI Search), AWS (EC2, S3), GCP, Supabase, Vercel, Prometheus, Grafana, Linux, Monitoring & Alerting

AI / ML: Machine Learning, Deep Learning, NLP, PyTorch, Scikit-Learn, XGBoost, GraphSAGE, PyTorch Geometric, Hugging Face, LangChain, OpenAI API, FAISS, Computer Vision, OpenCV, D3.js, Chart.js

Experience

AI / Software Engineering Intern, Provaantech Apr 2026 – August 2026
GitHub | Demo

- Worked with the product team to gather requirements and design an AI Candidate Discovery Engine that indexes and searches 110M+ records, returning top-10 ranked matches in under 10 seconds.
- Collaborated with backend engineers to architect and test a hybrid retrieval pipeline (OpenAI embeddings, Azure AI Search, BM25 ranking, semantic re-ranking), applying sound engineering principles to balance accuracy, cost, and latency.
- Wrote unit and integration tests for the retrieval and ranking modules, and integrated GPT-4o-mini for candidate-fit explanations; participated in code reviews and incorporated senior engineer feedback to improve output quality and explainability.

Software Engineering Intern, AVSS Tech Dec 2025 – Jan 2026

- Partnered with the product and hardware teams to build a real-time face recognition pipeline achieving 92%+ accuracy for enterprise access control, from requirements through testing and deployment.
- Debugged and improved system performance by reducing inference latency 35% via adaptive frame batching, frame skipping, and hardware-level optimizations, applying analytical rigor under deadline pressure.
- Designed a modular detection-and-verification pipeline with graceful fallback logic for edge-case inputs, and wrote test cases to drive consistency and reliability of the system in production.

Researcher, IEEE GRSS, Manipal Institute of Technology Jan 2025

- Collaborated with a research team to formulate the problem and engineer a two-stage hurdle stacking ensemble across 210 weather stations using 15 years of NASA POWER reanalysis data, presenting results for IGARSS 2026 publication.
- Achieved R^2=0.581, MAE=2.17 mm, Skill Score=0.510; enforced leakage-free chronological cross-validation with out-of-fold meta-learning, incorporating peer feedback and industry best practices into the methodology.

Projects

PII Redaction Microservice – Privacy Engineering Demo

- Designed and shipped a zero per-request-cost PII redaction service for resume PDFs using Microsoft Presidio, spaCy NER, and custom regex recognizers for region-specific formats (PAN, Aadhaar, +91 numbers).
- Deployed on Azure AKS via FastAPI, Docker, and Kubernetes; engineered a typed, zero-disk-I/O architecture with Pydantic v2 validation and wrote test coverage for high-throughput batch processing with Redis caching.
- Instrumented the service with Prometheus and Grafana to drive consistency in monitoring, observability, and operations at scale.

CrowdWisdom.AI – Multi-Agent LLM System GitHub | Live Demo

- Built, with a teammate, a multi-agent orchestration system for automated workflow execution, improving scheduling efficiency by 40% through algorithmic process design and testing.
- Integrated ElevenLabs Voice AI for real-time conversational interaction with dynamic LLM-driven agent routing across service boundaries.

Awards & Achievements

KisanMitra – Google Vibe2Ship Hackathon

- **Winner, Google Vibe2Ship Hackathon (Rs. 40,000 Prize).**
- Led a team to design, build, and ship KisanMitra, a functional product delivered within the hackathon's rapid build-and-ship format, collaborating closely with teammates under tight time constraints.

AgniSense – Predictive Fire Risk & Operational Command Platform GitHub

- **Top Winner, National-Level Hackathon (Rs. 75,000 Prize).**
- Worked with teammates to design a unified feature store integrating 5 municipal datasets, and built a FastAPI + Uvicorn ML backend processing 1.7+ GB telemetry for real-time risk-score inference at scale.

Parakh.AI – Edge-Native Industrial Quality Control

- **Top 3, National-Level Hackathon (Rs. 20,000 Prize).**
- Engineered, as part of a team, a zero-shot industrial anomaly detection pipeline using WideResNet-50, DINOv2, and FAISS similarity search, achieving 99.1% AUROC through iterative testing and tuning.
- Deployed on-device for real-time industrial quality control with no cloud dependency; used Mistral-7B to auto-generate defect explanations for end users.

SmurfPakad – Blockchain Fraud Detection via GNNs GitHub | Demo

- **Top 3, State-Level Hackathon (Rs. 20,000 Prize).**
- Collaborated with a cross-functional team to design and implement a GraphSAGE-based GNN detecting smurfing and money-laundering patterns across blockchain transaction graphs.
- Delivered a full-stack solution with FastAPI, PostgreSQL, Docker, and D3.js force-directed graph visualizations under a hackathon deadline.

Certifications

- **Develop Generative AI Applications: Get Started** – Coursera **View Certificate**